

2078

Bachelor of Science in Computer Science and Information Technology

Course Title: **Simulation and Modeling**

Code No: **CSC328** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: , taking **6±3 minutes for each**. Draw GPSS block diagram to simulate the above problem for 100 requisitions.12. Write short notes on (any two): a. Digital analog simulator b. Simulation tools

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define queuing system. Explain the Kendall's notation for queuing system. What are the various performance measures in single server queuing system? Explain which of them determine system stability and how?
2. Define true random numbers and pseudo random numbers with its properties. The sequence of numbers 0.64, 0.25, 0.58, 0.72, 0.90 has been generated. Use KS Test with $D\alpha=0.050 \Rightarrow 0.512$ to determine if the hypothesis that they are uniformly distributed on interval [0,1] can be rejected.
3. What do you understand by dynamic mathematical model? Explain with example. Differentiate it with static mathematical model.

Section B

Attempt any Eight questions[8×5=40]

4. Describe the phases in simulation.
5. Explain the concept of discrete event simulation. Explain poisson's arrival pattern.
6. Explain Monte Carlo simulation method with an example.
7. Define the terms verification, calibration, validation and accreditation of models
8. Use multiplicative congruential method to generate a sequence of random numbers with $x_0 = 7$, $a = 11$, $m = 16$.
9. Why is estimation methods used in simulation? Explain.
10. Explain the importance of elimination of initial bias during simulation.
11. Workers come to a supply store at the rate of one every 6 ± 2 minute. Their requisitions are processed by one of the two clerks who take 8 ± 2 minutes for each requisition. The requisitions are then passed to a single storekeeper who fills them one at a time, taking 6 ± 3 minutes for each. Draw GPSS block diagram to simulate the above problem for 100 requisitions.
12. Write short notes on (any two): a. Digital analog simulator b. Simulation tools